PEDAGOGY + AI · GRADES 6-12 · FREE CARD DECK

# Lesson Lens

Run the lesson you have already planned through a research lens. Your AI tool keeps your lesson and suggests at most two small, concrete upgrades.

Not sure? Start here: Formative check (hinge question, exit ticket). It fits almost any lesson: it shows you what students understood before you move on.

## Using the answer

Plan as usual
A few lines: what happens, in what order.

Read Keep first
It names what already works.

Try at most two upgrades
The Nudge is a question you can ignore.

Look for the Check
It tells you whether the change worked.

## The twelve lenses

1. Retrieval practice Strong
2. Spacing across the unit Strong
3. Cognitive load & worked examples Strong
4. Dual coding (words + visuals) Promising
5. Formative check (hinge question, exit ticket) Strong
6. Feedback that moves learning Strong
7. Explicit instruction (Rosenshine) Strong
8. Metacognition & self-regulation Strong
9. Dialogic discussion Promising
10. Belonging & safety to try Promising
11. Thinking depth (Bloom / SOLO) Framework
12. The trade-off question (AI use) Choosewise

## How strong is the evidence?

Strong Meta-analyses of classroom studies show that teaching this way improves learning.

Promising Controlled studies show gains, but results are mixed, small in number or dependent on context.

Framework A useful structure for planning, with no intervention studies behind it as a method.

Choosewise Choosewise's own question, not a research finding.

LENS 1

# Retrieval practice

Students pull what they have learned out of memory, instead of reading it again.

Strong

### History · Grade 9

**LESSON**

1. Starter: I recap last lesson on the causes of the First World War (5 min).
2. Slides on the July Crisis of 1914, students take notes.
3. Pairs read a source extract and answer three questions.
4. Class discussion: could the war have been avoided?
5. Students re-read their notes for homework.

**KEEP**

The source extract in pairs puts students to work on real evidence, and the closing discussion asks for a judgement, not just recall.

**UPGRADE**

1. Swap your recap for a five-question quiz from memory: name two long-term causes, what the alliance system was, and who was assassinated in Sarajevo. Students check against their notes only afterwards.
2. Replace re-reading for homework with three closed-book questions on the July Crisis, answered first and then checked against the notes.

**NUDGE**

Is re-reading notes at home doing what you hope, when answering from memory would make the same content stick longer?

**CHECK**

In the starter quiz, which cause most students could not name. That is what to revisit.

**For the AI.** Look for moments where students re-read, copy or listen to a recap, and where they could instead recall from memory without notes: a quick quiz, a brain dump, a closed-book question. A good upgrade replaces one recap with low-stakes recall of this lesson's key content, or of earlier content it builds on.

**Evidence.** Practice testing is one of two strategies rated high utility in the best-known review of study techniques (Dunlosky et al. 2013). A meta-analysis of classroom studies confirms it improves learning in school (Yang et al. 2021), across ages and subjects (Agarwal, Nunes & Blunt 2021).

LENS 2

# Spacing across the unit

Practice is spread out over days and weeks, so earlier content keeps coming back.

Strong

**Maths** · Grade 7

| | |
|---|---|
| LESSON | 1. Introduce adding fractions with different denominators.<br>2. Three worked examples on the board.<br>3. Worksheet: 25 questions on adding fractions.<br>4. Go through answers at the end.<br>5. Next week we start decimals. |
| KEEP | Worked examples before independent practice give students a clear model to follow. |
| UPGRADE | 1. Cut the worksheet to 12 questions today and save the rest as five-minute starters spread over the next three weeks, including during decimals.<br>2. Add two questions from earlier in the term (equivalent fractions, simplifying) to today's set, mixed in with the new ones. |
| NUDGE | If fractions disappear once decimals start, how much of today's practice will still be there at the end of term? |
| CHECK | In two weeks, whether a fraction starter goes as well as today's last questions did. |

**For the AI.** Look at where this lesson sits in the unit. Check whether all practice of today's content happens today, and whether content from earlier weeks returns at all. A good upgrade moves some practice to later lessons, or brings back one or two items from weeks ago in a short mixed set.

**Evidence.** Distributed practice is the other strategy rated high utility by Dunlosky et al. 2013. A 2022 review of classroom and lab research finds that spacing and retrieval together give the most durable learning (Carpenter, Pan & Butler 2022).

LENS 3

# Cognitive load & worked examples

New material is shown in small steps with worked examples, so working memory is not overloaded.

Strong

**Chemistry** · Grade 10

| | |
|---|---|
| LESSON | 1. Introduce moles, molar mass and Avogadro's number on one slide.<br>2. Students read the textbook page while I explain.<br>3. Worksheet: 10 mole calculations, mixed difficulty.<br>4. Help students who get stuck. |
| KEEP | The worksheet gives every student real calculation practice in the same lesson as the new idea. |
| UPGRADE | 1. Split the slide: introduce molar mass and one calculation first, and leave Avogadro's number for later in the lesson or the next lesson.<br>2. Before the worksheet, show two fully worked mole calculations, then one half-finished one students complete, and start the worksheet with the easiest questions. |
| NUDGE | Can students read the textbook and listen to you at the same time, or is one of them the better source in that moment? |
| CHECK | How many students can finish the half-finished example on their own before the worksheet starts. |

**For the AI.** Look for places where students meet a lot of new information at once: several new terms, busy slides, text and talk competing, or problems before any worked example. A good upgrade removes or splits one load point, or adds a worked example that fades into independent practice.

**Evidence.** Cognitive load theory rests on decades of replicated experiments. The worked-example effect is one of its most robust findings, confirmed in a 2023 meta-analysis, and it fades as students gain expertise (Sweller, van Merriënboer & Paas 2019; Barbieri et al. 2023).

LENS 4

# Dual coding (words + visuals)

Words and a clear visual explain the same idea together, without extra decoration.

Promising

**Geography** · Grade 8

**LESSON**

1. Slides on the water cycle: six bullet points per slide.
2. I read the slides aloud and add examples.
3. Students copy the key words into their books.
4. Video clip on flooding (4 min).
5. Exit question: explain the water cycle in your own words.

**KEEP**

Ending with an explanation in their own words shows you what students actually understood, not just what they copied.

**UPGRADE**

1. Replace the bullet slides with one blank water cycle diagram that you draw and label live while students label their own copy.
2. Drop the slide text you read aloud and keep only the diagram on screen, so students listen and look rather than read and listen.

**NUDGE**

Is copying key words doing what labelling a diagram would do, when students must connect the stages themselves?

**CHECK**

Whether students' exit explanations follow the diagram's order of stages, and which link between stages is missing most.

**For the AI.** Look at how slides, diagrams and talk work together. Check for text read aloud from slides, decorative images, or ideas that a simple diagram, timeline or map would make clearer. A good upgrade replaces one text-heavy part with a visual that students build or label while the teacher explains.

**Evidence.** An overview of meta-analyses on multimedia design finds reliable benefits from pairing words with relevant visuals and removing extras, mostly in short studies (Noetel et al. 2022). Long-term classroom evidence is thinner.

LENS 5

# Formative check (hinge question, exit ticket)

A quick check during or at the end of the lesson shows what students understood, so the next step is based on evidence.

Strong

**Science** · Grade 8

**LESSON**

1. Starter: what do plants need to grow?
2. Demonstration: pondweed in water under a lamp, bubbles of oxygen.
3. Slides with three diagrams of photosynthesis.
4. Worksheet on the word equation.
5. Students summarise the chapter.

**KEEP**

The demonstration gives students something concrete to reason about before they write, and the worksheet on the word equation gives you evidence you can check.

**UPGRADE**

1. Before the worksheet, ask a hinge question: "Where does most of a tree's dry mass (the wood, once the water is dried out) come from? A soil, B water, C air, D sunlight." Everyone shows a letter at once. If more than a quarter choose anything other than C, re-teach with the pondweed before moving on.
2. Replace the chapter summary with an exit ticket: write the word equation, and name one thing a plant takes in and one thing it gives off. Sort the tickets into three piles (secure, partly, not yet) and open tomorrow's lesson with the gap in the "not yet" pile.

**NUDGE**

Right now the first time you see what students understood is the worksheet. Would you rather find out while there is still time to re-teach?

**CHECK**

How many choose C on the hinge question, and how many exit tickets end up in the "not yet" pile.

**For the AI.** Look for the point where the teacher moves on without knowing whether students understood. A good upgrade adds one hinge question at that point (multiple choice, with wrong options built on common misconceptions, answered by everyone at once) or a short exit ticket, and says what the teacher does with each result.

**Evidence.** Black & Wiliam's 1998 review made the case that formative assessment raises achievement, and a 2024 meta-analysis of 118 K-12 studies confirms a positive, more modest average effect (Yao et al. 2024). A whole-school EEF trial found a similar, real effect at scale.

LENS 6

# Feedback that moves learning

Feedback tells students where they are going, how they are doing and what to do next, and they act on it.

Strong

**English** · Grade 11

| | |
|---|---|
| LESSON | 1. Return marked essays on Of Mice and Men with grades and comments.<br>2. Students read their comments.<br>3. Introduce the next essay question.<br>4. Students start planning the new essay. |
| KEEP | Moving straight from one essay to planning the next gives students a natural place to use what they learned. |
| UPGRADE | 1. Before students see the grade, give them 15 minutes to rewrite one paragraph using the comment on it. Show the grade afterwards.<br>2. Pick the most common weakness across the class (for example, quotes without analysis), model one improved paragraph, and have students find and fix one example in their own essay. |
| NUDGE | When the grade and the comments arrive together, which one do you think students read? |
| CHECK | Whether the rewritten paragraphs show the change the comment asked for, and whether it carries over into the new plan. |

**For the AI.** Look at when and how students get feedback in this lesson, and whether they get time to use it. A good upgrade makes feedback specific to the task, points to the next step, and builds in time in class for students to act on it. It never adds more written comments for the teacher to write.

Evidence. Feedback is among the best-evidenced approaches in the EEF Teaching and Learning Toolkit. A 2020 meta-analysis finds a medium average effect with large variation: feedback that carries rich information about the task and how to improve works much better than praise, rewards or punishment (Wisniewski, Zierer & Hattie 2020).

LENS 7

# Explicit instruction (Rosenshine)

The teacher models, guides practice and checks understanding before students work on their own.

Strong

**Maths** · Grade 6

| | |
|---|---|
| LESSON | 1. Introduce finding the area of a triangle.<br>2. Show the formula on a slide.<br>3. Students work through textbook exercises alone.<br>4. I walk around and help.<br>5. Check answers at the end. |
| KEEP | Walking around during practice lets you see who is stuck while there is still time to help. |
| UPGRADE | 1. Before the formula, cut a rectangle in half along its diagonal and think aloud: why is the triangle half the rectangle? Then draw a rectangle around a triangle that is not right-angled and show that it is still half. Model two examples step by step.<br>2. Add a guided round: three triangles on mini whiteboards, everyone shows the answer, and move to the textbook only when most get them right. |
| NUDGE | If students go from the formula straight to working alone, how do you know they are ready before half the lesson has gone? |
| CHECK | The share of correct whiteboards in the guided round. Aim for most of the class before independent work. |

**For the AI.** Look at the sequence from new content to independent work: is there modelling, guided practice with checks, and a high success rate before students work alone? A good upgrade adds or strengthens one missing step, such as thinking aloud while modelling, or a short guided round with whiteboards before independent work.

**Evidence.** A meta-analysis of 328 studies over half a century found consistently positive effects of Direct Instruction, a highly structured form of explicit teaching (Stockard et al. 2018). Rosenshine's principles summarise the wider research on explicit instruction for classroom use.

LENS 8

# Metacognition & self-regulation

Students plan, monitor and evaluate their own learning, and the teacher shows them how.

Strong

**Spanish** · Grade 9

**LESSON**

1. Vocabulary for describing your town.
2. Listening exercise from the textbook audio.
3. Students write a short paragraph about their own town.
4. Swap with a partner and read each other's paragraph.

**KEEP**

Writing about their own town makes the vocabulary personal, and reading a partner's text gives a real reader.

**UPGRADE**

1. Before the listening exercise, ask: "What will you listen for, and what will you do when you miss a word?" Model your own strategy once: listen for the words you know and guess from context.
2. Before the paragraph, students write one line on how they will check their own text (verb endings, adjective agreement), then use it before swapping.

**NUDGE**

Students check each other's paragraphs. Would they learn more by checking their own first, with a list they made themselves?

**CHECK**

Whether students can name the strategy they used when they missed a word in the listening.

**For the AI.** Look for moments where students could plan how to tackle a task, check how it is going, or look back on what worked, and where the teacher could model that thinking aloud. A good upgrade adds one short planning or reflection step tied to this lesson's task, not a general reflection form.

Evidence. Metacognition and self-regulation is one of the highest-impact, best-evidenced approaches in the EEF Teaching and Learning Toolkit (+8 months, 4 padlocks), when it is taught explicitly inside subject content.

LENS 9

# Dialogic discussion

Students build on each other's ideas in discussion, and the teacher's questions push for reasons.

Promising

**Religion and ethics** · Grade 10

| | |
|---|---|
| LESSON | 1. Introduce the question: is it ever right to break the law?<br>2. Short text on civil disobedience.<br>3. Whole-class discussion, students put up their hands.<br>4. Students write their own answer. |
| KEEP | The question has no single right answer, which is exactly what a real discussion needs. |
| UPGRADE | 1. Give one minute of silent thinking and one minute in pairs before anyone speaks to the whole class, so more students have something to say.<br>2. During the discussion, ask the next speaker to start with "I agree with … because" or "I see it differently from … because", so students answer each other, not just you. |
| NUDGE | With hands up, the same few students often do most of the talking. Who would speak if you chose speakers after pair talk instead? |
| CHECK | How many different students speak, and how many responses refer to a classmate's point. |

**For the AI.** Look at the discussion parts of the lesson: who talks, what kind of questions are asked, and whether students respond to each other or only to the teacher. A good upgrade adds thinking time, an open question with more than one reasonable answer, or a move that makes students build on or challenge a classmate's point.

Evidence. An EEF trial of Dialogic Teaching found positive effects in English, science and maths, though a follow-up found no effect on later national tests. A large classroom study links participation and elaborated reasoning in whole-class talk to better outcomes (Howe et al. 2019). Results vary with how well it is done.

LENS 10

# Belonging & safety to try

Students feel they belong and that mistakes are part of learning, so they dare to try.

Promising

**PE and health** · Grade 7

| | |
|---|---|
| LESSON | 1. Warm-up.<br>2. Introduce the forward roll, I demonstrate.<br>3. Students take turns rolling in front of the class.<br>4. Best rolls shown as examples.<br>5. Cool-down. |
| KEEP | Your demonstration first gives everyone a clear picture of the movement before they try it. |
| UPGRADE | 1. Let students practise in groups of three on separate mats, with one roller and two helpers giving one tip each, instead of rolling one at a time in front of the class.<br>2. Show a common mistake (a hand placed wrong) and how to fix it, as well as a good roll, so getting it wrong becomes part of the lesson. |
| NUDGE | Rolling in front of the whole class can stop the least confident from trying at all. What would change if the first attempts happened out of sight of the group? |
| CHECK | How many students get several attempts, and whether the hesitant ones try at all. |

**For the AI.** Look for moments where students risk looking wrong in front of others, are picked on the spot, or compare themselves publicly. A good upgrade lowers the social risk while keeping the challenge: thinking time, practice in pairs or small groups first, or treating a common mistake as useful for the whole class.

**Evidence.** A meta-analytic review links school belonging to motivation, well-being and achievement in secondary school (Korpershoek et al. 2020). Brief belonging interventions have helped some students, but a 2023 study of almost 27,000 students found they worked only where the school gave students real chances to belong (Walton et al. 2023).

LENS 11

# Thinking depth (Bloom / SOLO)

Questions and tasks move from recalling facts to explaining, applying and judging.

Framework

**Social studies** · Grade 11

| | |
|---|---|
| LESSON | 1. Introduce the three branches of government.<br>2. Students fill in a table: branch, who, what they do.<br>3. Quiz: match the branch to its role.<br>4. Homework: read the next chapter. |
| KEEP | The table gives students a clear structure for content they need before they can reason about it. |
| UPGRADE | 1. Replace the matching quiz with one case: "The government proposes a new law, but a majority in parliament votes against it. What can each branch do next?" Students answer in pairs.<br>2. Add a final question that asks for a judgement: "Which branch do you think has the most power in our country, and what is your strongest reason?" |
| NUDGE | If every task asks students to recall who does what, when do they get to use it? |
| CHECK | Whether the answers to the case explain how the branches check each other, or only list them again. |

**For the AI.** Look at the questions and tasks in the lesson and what depth of thinking they ask for. Check whether students only recall and describe, or also explain, compare, apply and judge. A good upgrade turns one recall task into a deeper one on the same content, without adding a new activity.

**Evidence.** Bloom's revised taxonomy (Krathwohl 2002) and the SOLO taxonomy (Biggs & Collis 1982) are widely used to plan and judge the depth of questions and tasks. They are languages for planning, not methods tested in intervention studies.

LENS 12

# The trade-off question (AI use)

What do students get worse at if AI does this part?

Choosewise

**English** · Grade 10

| | |
|---|---|
| LESSON | 1. Read a short story in class.<br>2. Discuss the main character's choices.<br>3. Introduce the essay question: how does the writer build tension?<br>4. Homework: write the essay (600 words). |
| KEEP | Reading and discussing the story together gives every student their own ideas to start from before writing. |
| UPGRADE | 1. Move the planning into class: in the last 15 minutes, students pick three moments of tension and write one sentence each on how the writer builds it. That plan is their own thinking, done before anyone opens an AI tool.<br>2. Ask students to hand in the plan with the essay and to highlight one sentence they rewrote and why, so their process shows. |
| NUDGE | Essay homework is exactly the part many students hand to AI. If AI writes the analysis, what do they get worse at: spotting how tension is built, or putting it into words? |
| CHECK | Whether the in-class plans name specific moments in the text, and whether the essays build on them. |

**For the AI.** Apply this lens even if the lesson has no AI in it. Find the part of the lesson or homework that students are most likely to hand over to AI on their own, name the thinking that part is meant to train, and say what students get worse at if AI does it. If AI is already planned, judge it the same way. A good upgrade protects that thinking, for example by moving it into class, making students' own process visible, or using AI only after students have tried.

**Evidence.** This is Choosewise's own question, not a research finding. It draws on research showing that unrestricted AI help can raise practice scores while lowering learning once the help is removed (Bastani et al. 2025).

# Your own prompt

Fill in the blanks, then type the prompt into any AI tool: Copilot, Gemini, ChatGPT or Claude. Leave out the line "Students already know" if you have nothing to add. Check the suggestions before the lesson. The AI proposes; you decide.

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Do not include student names or student work in this conversation.

You are helping a Grade _____ _____ teacher improve a lesson they
have already planned. Keep their lesson and their way of teaching. Do not rewrite the
lesson.

Lens: _____ (the lens's name and one-line description
from its card)
How to apply it: _____ (the "For the AI" text from its card)

The lesson as planned:
_____
_____
_____

Students already know: _____

Reply in plain text, no tables, with exactly these four parts:
KEEP: one or two lines on what already works in this lesson and why.
UPGRADE: at most two changes in total, numbered. Each must use this lesson's own content
and be small enough to make this evening.
NUDGE: one question, only if the research behind the lens points to a clearly stronger
option than what is planned. Otherwise write "None".
CHECK: what the teacher should look for in class to see whether it worked.
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Faster online: the prompt builder at choosewise.education/lesson-lens fills this in for you, and lets you add a second lens. If students use AI themselves, check the tool's age rules, your national rules and your school policy first.